# MATH 150: WEIL CONJECTURES

SAIR SHAIKH

## 1. Motivation:

I will begin by stating the Weil Conjectures (there before class):

**Theorem 1.1.** *If $X$ is a non-singular, geometrically connected, $n$-dimensional projective variety over $\mathbb{F}_q$. Define:*

$$Z(t) = \exp\left(\sum_{m\geq 1} \frac{N_m}{m} t^m\right)$$

*Then, the following hold:*

(1) *(Rationality) $Z(t)$ is a rational function, i.e. $Z(t) \in \mathbb{Q}(t)$.*
(2) *(Functional Equation) If $E$ is the Euler characteristic of $X$, then:*

$$Z(\frac{1}{q^n t}) = \pm q^{nE/2} t^E Z(t)$$

(3) *(Riemann Hypothesis) One can write:*

$$Z(X,t) = \frac{P_1(t)\cdot P_3(t)\cdot P_{2n-1}(t)}{P_0(t)\cdot P_2(t)\cdot P_{2n}(t)}$$

*with $P_0(t) = 1-t$, $P_{2n}(t) = 1-q^n t$ and for $1 \leq 2n-1$, we have $P_i(t) \in \mathbb{Z}[t]$,*

$$P_i(t) = \prod_j (1-\alpha_{i,j} t)$$

*with $|\alpha_{i,j}| = q^{i/2}$.*

(4) *(Comparison Theorem) If $X$ come from a good reduction mod $p$ of a non-singular projective variety $Y$ defined over a number field in $\mathbb{C}$, then the degree of $P_i$ is the $i$th Betti number of $Y$.*

**Remark 1.2.** *A priori, we do not have a defined Euler characteristic. But once we have rationality, we can define it as the degree of the denominator minus the degree of the numerator.*

**Remark 1.3.** *Weil proved these conjectures for curves of arbitrary genus in 1948 and for diagonal hypersurfaces in 1949. This is why he conjectured them.*

In this talk, we will reduce the problem of proving the first two of these to finding a cohomology theory satisfying some axioms, called a Weil Cohomology theory.

To get a hold of the zeta function, we really need to understand $X(\mathbb{F}_q^r)$. Recall, from Henry's talk (initial object in the category of talks),

**Theorem 1.4.** *Let $X$ be a non-singular projective variety over $\mathbb{F}_q$. We have the absolute frobenius* $\mathrm{Frob}_q : X \to X$ *given by* $\mathrm{id}$ *on $X$ and $u \mapsto u^q$ on $\mathcal{O}_X$. Then, $X(\mathbb{F}_{q^r})$ is in natural bijection with $X(\overline{\mathbb{F}}_q)$ fixed by* $\mathrm{Frob}_q^r$.
*More geometrically, define $\overline{X} = X \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$ and $F_q := \mathrm{Frob}_q \times \mathrm{Id}$ on $\overline{X}$. Let $\Delta, \Gamma_r \subset \overline{X} \times \overline{X}$ be the diagonal and the graph of $F^r$, respectively. Then $X(\mathbb{F}_q^r)$ is in bijection with the closed points of $\Gamma_r \cap \Delta$.*

Thus, we need a way to count the closed points of $\Gamma_r \cap \Delta$. To this end, we do have a result from classical topology, the Lefschetz fixed-point theorem (from Ribhu's talk):

**Theorem 1.5.** *Let $M$ be an $n$-dimensional oriented compact manifold without boundary and $f : M \to M$ a map with isolated fixed points. Then,*

$$(\Gamma_f \cdot \Delta) = \sum_{i=0}^{n} (-1)^i \operatorname{Tr}(f_i^* | H^i)$$

We recall from Ribhu's talk that the proof of this required a few key ingredients:

(1) We converted the problem of counting the intersection number to linear algebra. In particular, for submanifolds $S, N \subset M$, we found forms $\eta_S, \eta_N$ such that:

$$(S \cdot N) = \int_M \eta_S \wedge \eta_N$$

(2) We used some nice properties of the cohomology theory, such as Poincare duality and the Kunneth formula to derive the result.

We want to do something similar in the algebraic geometry setting, where intersection theory is more complicated. So we start by reviewing some properties of intersection theory.

## 2. Intersection Theory Interlude:

Let $X$ be a smooth variety over $k$.

**Definition 2.1.** *(1) Define the groups $Z_r(X)$ are free abelian group generated by closed, irreducible subvarieties of dimension $r$. The elements of $Z_r(X)$ are called algebraic "$r$-cycles".*

*(2) Let $V$ be a $(k+1)$ dimensional subvariety of $X \times \mathbb{P}^1$ such that $f : V \to \mathbb{P}^1$ is dominant. For $P \in \mathbb{P}^1$ rational over the ground field, the fiber $f^{-1}(P)$ is isomorphic to a subvariety of $X$. Denote this as $V(P)$. Rational equivalence is the equivalence relation generated by setting:*

$$[V(0)] - [V(\infty)] \sim 0$$

*(3) Define the Chow groups $A_r(X)$ are the quotient groups under the rational equivalence relation $\sim_{rat}$. For $Z \in Z_r(X)$, $[Z] \in A_r(X)$ is the class of $Z$. Set $A^i = A_{n-i}$.*

*(4) There is an intersection product making $A^*(X)$ into a graded ring. This is difficult to define in generality, but is related to the intuitive intersection of subvarieties. For $W \in A^i(X)$ and $Z \in A^{n-i}(X)$,*

$$[W \cap Z] = \sum_{P \in W \cap Z} l(\mathcal{O}_{W \cap Z, P})[P]$$

From morphisms on varieties, we have two fundamental morphisms induced on the Chow groups:

**Definition 2.2** (Proper pushforward)**.** *Let $f : X \to Y$ be a proper morphism. For a closed subvariety, $V \subset X$ of dimension $r$, we define:*

$$f_* : Z_r(X) \to Z_r(Y), \qquad f_*(V) = \deg(V/f(V))[f(V)]$$

*where $\deg(V/f(V))$ is the degree of the extension $[K(V) : K(f(V))]$ if $\dim(f(V)) = \dim(V)$ and $0$ otherwise.*

**Remark 2.3.** *We need properness so that fibers are compact and $\deg(V/f(V))$ is well-defined.*

**Theorem 2.4.** *The pushforward has the following properties:*

*(1) The pushforward induces a well-defined map $A_r(X) \to A_r(Y)$.*
*(2) It is (covariant) functorial.*
*(3) For $\alpha = \sum_P n_P[P]$ a 0-cycle, define the degree of $\alpha$ by $\deg(\alpha) = \sum_P n_P$. Then, $\deg(\alpha) = f_*(\alpha)$ where $f : X \to \operatorname{Spec} k$.*

**Definition 2.5** (Flat Pullback)**.** *Let $f : X \to Y$ be a flat morphism of relative dimension $r$. Define the pull-back map:*

$$f^* : Z_i(Y) \to Z_{i+r}, \qquad f^*([V]) = [f^{-1}([V])]$$

**Remark 2.6.** *We need flat so that the fibers are well-behaved.*

**Theorem 2.7.** *The pullback has the following properties:*

*(1) The pushforward induces a well-defined map $A_r(X) \to A_r(Y)$.*
*(2) It is (contravariant) functorial.*

## 3. Weil Cohomology Theories:

**Definition 3.1.** *(Weil Cohomology Theory)*
*A Weil Cohomology theory with coefficients in a char 0 field $K$ consists of the following data:*

1. *A contravariant functor $X \to H^*(X) = \bigoplus_i H^i(X)$ from the category of smooth, connected, projective varieties over (algebraically closed) $k$ to the category of graded commutative $K$-algebras. The product of $\alpha, \beta \in H^*(X)$ is denoted $\alpha \cup \beta$. We write $f^* := H(f)$.*
2. *A linear map $\mathrm{Tr} = \mathrm{Tr}_X : H^{2\dim(X)}(X) \to K$.*
3. *The cycle class map: $\mathrm{cl} : Z^r(X) \to H^{2r}(X)$ mapping each closed irreducible subvariety $Z \subseteq X$ of codimension $r$ to a cohomology class $\mathrm{cl}(Z)$.*

*On this data, we require the following:*

1. *(Fin. dim.) All $H^i(X)$ are finite dimensional vector spaces over $K$. Moreover, $H^i(X) = 0$ except when $0 \leq i \leq 2\dim(X)$.*
2. *(Point) If $X = \operatorname{Spec} k$, then $\mathrm{cl}(X) = 1$ and $\mathrm{Tr}_X(1) = 1$.*
3. *(Kunneth property) For $X, Y$, if we define $p_X : X \times Y \to X$ and $p_Y : X \times Y \to Y$ to be the projections, then the $K$-algebra homomorphism:*
$$H^*(X) \otimes_K H^*(Y) \to H^*(X \times Y), \quad \alpha \otimes \beta \mapsto p_X^*(\alpha) \cup p_Y^*(\beta)$$
*is an isomorphism.*
4. *(Poincaré duality) We have $\mathrm{Tr} : H^{2\dim(X)}(X) \cong K$, and for every $0 \leq i \leq 2\dim(X)$, the bilinear map:*
$$H^i(X) \otimes_K H^{2\dim(X)-i}(X) \to K, \quad (\alpha, \beta) \mapsto \mathrm{Tr}_X(\alpha \cup \beta)$$
*is a perfect pairing.*
5. *(Exterior product) For varieties $X, Y$ and every closed irreducible varieties $Z \subseteq X$ and $W \subseteq Y$, we have:*
$$\mathrm{cl}(Z \times W) = p_X^*(\mathrm{cl}(Z)) \cup p_Y^*(\mathrm{cl}(W))$$
*and further if $\mathrm{cl}(Z) \in H^{2\dim(X)}(X)$ and $\mathrm{cl}(W) \in H^{2\dim(Y)}(Y)$ then:*
$$\mathrm{Tr}_{X \times Y}(\mathrm{cl}(Z \times Y)) = \mathrm{Tr}_X(\mathrm{cl}(Z)) \mathrm{Tr}_Y(\mathrm{cl}(W))$$
6. *(Pushforward) For every (proper) morphism $f : X \to Y$ and for every irreducible closed subvariety $Z \subseteq X$, we have for every $\beta \in H^{2\dim(Z)}(X)$:*
$$\mathrm{Tr}_X(\mathrm{cl}(Z) \cup f^*(\beta)) = \mathrm{Tr}_Y(\mathrm{cl}(f_*(Z)) \cup \beta)$$
7. *(Pullback) Let $f : X \to Y$ be flat of relative dimension $r$ and $Z \subseteq Y$ a closed irreducible subvariety with $[f^{-1}(Z)] = \sum_{i=1}^r m_i W_i$, then $f^*(\mathrm{cl}(Z)) = \sum_{i=1}^r m_i \mathrm{cl}(W_i)$. That is, $f^*(\mathrm{cl}(Z)) = \mathrm{cl}(f^*(Z))$.*

We define the pushforward map by using the Poincare duality:

**Definition 3.2.** *(Pushforward) For $f : X \to Y$ a morphism and $\alpha \in H^i(X)$, define $f_*(\alpha) \in H^{i-(2m-2n)}(Y)$ to be:*

$$\mathrm{Tr}_Y(f_*(\alpha) \cup \beta) = \mathrm{Tr}_X(\alpha \cup f^*(\beta))$$

*for all $\beta \in H^{2m-i}(X)$.*
*That is, using axiom 6, $f_*(\mathrm{cl}(Z)) = \mathrm{cl}(f_*(Z))$.*

## 4. Grothendieck-Lefschetz Trace Formula:

Given such a cohomology theory, we get the following theorems as a formal consequence:

**Theorem 4.1.** *For $X$ non-singular, connected, projective variety:*

*(1)* $\mathrm{cl} : A^*(X) \to H^{2*}(X)$ *is a ring map*
*(2)* $f^*$ *and* $f_*$ *are compute with this map.*
*(3) For* $\alpha_i \in A^{m_i}$ *for* $1 \leq i \leq r$ *such that* $\sum_i m_i = \dim(X)$*, then*

$$(\alpha_1 \cdot \ldots \alpha_r) = \mathrm{Tr}(\mathrm{cl}(\alpha_1) \cup \cdots \cup \mathrm{cl}(\alpha_r))$$

Additionally, using these axioms we can prove exactly what we want:

**Theorem 4.2.** *(Grothendieck-Lefschetz Trace Formula) If* $\phi : X \to X$ *is an endomorphism of the non-singular, connected, projective variety* $X$*, and if* $\Gamma_\phi, \Delta \subseteq X \times X$ *are the graph of* $X \times X$ *are the graph of* $\phi$ *and the diagonal, then:*

$$(\Gamma_\phi \cdot \Delta) = \sum_{i=0}^{2n} (-1)^i \,\mathrm{tr}(\phi^* | H^i(X))$$

*In particular, if* $\Gamma_\phi$ *and* $\Delta$ *intersect transversely, then the above expression computes* $|\{x \in X | \phi(x) = x\}|$*.*

*Proof.* Sketch of proof. The proof follows analogously to Ribhu's.

(1) First use (3) above to translate the problem into cohomology.
(2) Use Kunneth to express cohomology classes in $H^*(X \times X)$ as pullbacks of those coming from $H(X)$.
(3) Various pullback and pushforward gymnastics to write things in terms of a pullbacks of a nice basis wrt Poincare duality.
(4) Use pushforward+product axiom and collect terms

□

## 5. Proof of Weil Conjecture I and II

Given the Grothendieck-Lefschetz fixed point theorem, the proof of the rationality is reduced to linear algebra.

**Theorem 5.1.** *$Z(t)$ is a rational function, i.e. $Z(t) \in \mathbb{Q}(t)$*

*Proof.* First, note that:

$$N_m = (\Gamma_{F^m} \cdot \Delta)$$

where $\Gamma_{F^m}, \Delta \subseteq \overline{X} \times \overline{X}$. Via the GLFPT, we have:

$$N_m = \sum_{i=0}^{2n} (-1)^i \operatorname{tr}(F^{*m} | H^i(\overline{X}))$$

Thus,

$$Z(X,t) = \exp\left( \sum_{m \geq 1} \sum_{i=0}^{2n} (-1)^i \operatorname{tr}(F^{*m} | H^i(\overline{X})) \frac{t^m}{m} \right)$$

□

Then, we recall the following lemma from the very first talk.

**Lemma 5.2.** *For every endomorphism $\phi$ of a finite dimensional vector space $V$ over a field $K$, we have:*

$$\det(\mathrm{Id} - t\phi) = \exp\left( - \sum_{m \geq 1} \operatorname{tr}(\phi^m | V) \frac{t^m}{m} \right)$$

*Proof.* Michaela showed parts of this in the very first talk. This is easy.
Idea is to diagonalize $\phi$, rearrange, notice the Taylor expansion of log and conclude. □

*Proof.* Proof of Theorem. Continuing, we get:

$$Z(X,t) = \prod_{i=0}^{2n} \det(\mathrm{Id} - tF^* | H^i(\overline{X}))^{(-1)^{i+1}}$$

Thus, we conclude that $Z(X,t) \in K(t)$. We also know that $Z(X,t) \in \mathbb{Q}[[t]]$.
This implies $Z(X,t) \in \mathbb{Q}(t)$. This is an easy exercise in linear algebra. □

We have the first Weil Conjecture (modulo Henry and Semir defining $l$-adic cohomology).

Now, we turn our attention to the functional equation.

**Theorem 5.3.** *If $E$ is the Euler characteristic of $X$, then:*

$$Z\left(X, \frac{1}{q^n t}\right) = \pm q^{nE/2} t^E Z(t)$$

To handle this, the key step is another linear algebra lemma:

**Lemma 5.4.** *Let $\phi : V \times W \to K$ be a perfect pairing of vector spaces of dimension $r$ over the field $K$. If $\lambda \in K \setminus \{0\}$ and $f \in \mathrm{End}_K(V)$ and $g \in \mathrm{End}_K(W)$ are such that $\phi(f(v), g(w)) = \lambda\phi(v, w)$ for every $v \in V, w \in W$, then:*

$$\det(\mathrm{Id} - tg|W) = \frac{(-\lambda t)^r}{\det(f|V)} \det(\mathrm{Id} - \lambda^{-1} t^{-1} f|V)$$

*and*

$$\det(g|W) = \frac{\lambda^r}{\det(f|V)}$$

*Proof.* Mustata, lemma 4.15 or Hartshorne Appendix C, Lemma 4.3. □

*Proof of Theorem.* We want to apply the lemma to the Poincare pairing:

$$\phi_i : H^i(\overline{X}) \otimes H^{2n-i}(\overline{X}) \to H^{2n}(\overline{X}) \to K$$

defined by $\phi_i(\alpha, \beta) = \mathrm{Tr}(\alpha \cup \beta)$ and $f = F^*$ and $g = F^*$. Before that, we need to show there exists $\lambda \in K$ such that:

$$\phi_i(F^*(\alpha), F^*(\beta)) = \lambda\phi_i(\alpha, \beta)$$

First notice that:

$$\phi_i(F^*(\alpha), F^*(\beta)) = \mathrm{Tr}_{\overline{X}}(F^*(\alpha \cup \beta))$$

Thus, we need to show $F^*$ acts on $H^{2n}(\overline{X})$ by multiplication.
First, we notice that $F$ is a finite morphism of degree $q^n$.
Thus, for a point $Q \in \overline{X}$, $\deg([F^*Q]) = q^n$. We conclude by Axiom 7.

$$\phi_i(F^*(\alpha), F^*(\beta)) = q^n \,\mathrm{Tr}_{\overline{X}}(\alpha \cup \beta)) = q^n \phi_i(\alpha, \beta)$$

Letting $P_i(t) = \det(\mathrm{Id} - tF^*|H^i(\overline{X}))$ and $B_i = \dim_K H^i(\overline{X})$, we use the lemma to conclude:

$$\det(F^*|H^{2n-i}(\overline{X})) = \frac{q^{nB_i}}{\det(F^*|H^i(\overline{X}))}$$

and

$$P_{2n-i}(t) = \frac{(-1)^{B_i} q^{nB_i} t^{B_i}}{\det(F^*|H^i(\overline{X}))} P_i(1/q^n t)$$

This gives us:

$$\begin{aligned}
Z\left(X, \frac{1}{q^n t}\right) &= \prod_{i=0}^{2n} P_i\left(\frac{1}{q^n t}\right)^{(-1)^{i+1}} \\
&= \prod_{i=0}^{2n} P_{2n-i}(t)^{(-1)^{i+1}} \cdot \frac{(-1)^E q^{nE} t^E}{\prod_{i=0}^{2n} \det(F^* | H^i(\overline{X}))^{(-1)^i}} \\
&= \pm Z(X,t) \cdot \frac{q^{nE} t^E}{q^{nE/2}} \\
&= \pm q^{nE} t^E Z(X,t)
\end{aligned}$$

$\square$

We are done!